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Quantum Computing for Fraud Detection in Real-Time Payment Systems

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***Abstract*—The exponential growth of real-time payment (RTP) systems has agitated on the urgency of robust and efficient fraud detection mechanisms. Traditional machine-learning and rule-based systems often face challenges in processing large, very fast-moving transactional data, which results in delayed detection and substantial financial losses. Quantum computing, with its potential to perform parallel computations on complex probability spaces, offers a novel paradigm to enhance fraud detection. This paper centers on the application of quantum machine learning (QML) techniques—specifically Quantum Graph Neural Networks (QGNNs) and Quantum Support Vector Machines (QSVMs)—over RTP networks for detecting fraudulent activities. This paper will practically implement these hybrid classical and quantum model approaches using Qiskit along with Amazon Braket for datasets such as the IEEE-CIS credit card fraud dataset and quantum synthetically created transaction logs. These models will be compared with classical models in terms of accuracy, precision, recall and execution time. The results demonstrate that quantum-enhanced models offer robust detection accuracy and reduced computational latency, far more effectively when faced with severely imbalanced and nonlinear datasets. Specifically, this study buttresses the quantum forensic financial discipline as it brings to light the reality that quantum computing does not just stay in theoretical promise but can indeed be in practical advantages for scalable, real-time systems for the detection of frauds.**

***Index Terms*—Quantum Computing, Fraud Detection, Real-Time Payments, Quantum Machine Learning, Quantum Graph Neural Networks, Financial Cybersecurity, Qiskit, Amazon Braket, Quantum SVM, Transaction Anomaly Detection.**

I. Introduction

Real-time Payment Systems (RTP) have drastically changed the global financial system by offering instant money transfers in domestic and international setups. These RTP networks include entities like the RTP network in the United States, the Unified Payments Interface (UPI) in India, and the TARGET Instant Payment Settlements (TIPS) system within the European Union that have recently seen a rise in transactions, with a staggering figure of 70 plus billion RTP transactions structured worldwide only in 2023 [13].

However, all the speed comes with an imperial vulnerability. Money laundering for scam has steadily increased 46 percent in just two years through RTP, maintaining its claim as the fastest way to have the cash moved from anywhere in the U.S. immediately. By means of similar massive change frauds, crooks range from phishing to identity theft, legitimate accounts, and redirecting real-time transactions. The distinction lies in making fraudulent activities unknown to people in milliseconds- a technical feat beyond the current abilities of normal fraud-detection applications [1].

TABLE I
Common Types of Fraud in RTP Systems

Fraud Type	Description	Prevalence (%)
Phishing Attacks	Deceptive messages to gain access credentials	28
Synthetic Identity	Use of fictitious IDs to create new accounts	22
Account Takeover	Unauthorized access to user accounts	19
Friendly Fraud	Legitimate cardholder disputes a real transaction	17
Real-Time Redirection	Interception of transactions in progress	14

Source: [Statista, 2023] [6]

Classical fraud models such as rule engines, decision trees, and neural networks give reactiveness in their action, along with a high amount of post-processing required. Though their results show a large number of cases of false positives, they generally lack balanced input data. In such cases, the number of fraudulent transactions is so low as compared with nonfraudulent ones that results in biases in their training [4]. It is observed that these systems ask for significantly large processing power overshadowing them once the number of transactions to be scanned gets bigger and their dimensionality becomes significantly higher [11].

The paper considers a different proposal: quantum computing (QC), a variation that uses qubits allowing bit construction in simultaneous superposition states of 0 and 1 together. As evidenced by its characteristic, together with entanglement and

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quantum tunneling, QC can simply travel through a vast number of possible solutions simultaneously [5]. Toward the same aim, researchers propose quantum machine learning (QML) models such as quantum graph neural networks (QGNN) and quantum support vector machines (QSVMs) as the way forward to making the right decision at a faster pace [18], [3].

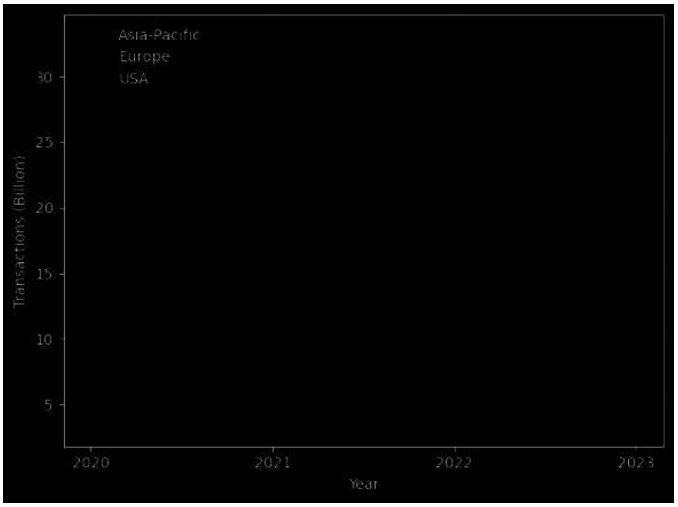

Fig. 1. Real-Time Payment Volume Growth by Region (2020–2023)
Source: Adapted from [McKinsey Payments Report, 2024] [17]

The application of quantum models has shown, in fact, some prototypes that are going to be generalized in fraud detection. For example, Deloitte Italy successfully utilized a hybrid quantum-classical detection pipeline using Amazon Braket, thus reducing model complexity to improve the detection rate [22]. Further Quantum Graph Neural Networks are also on the way that has shown to hold promise for flagship anomalies in transaction networks in a dataset that has a very complicated relationship with one another [8], [15].

Fig. 2. Classical Fraud Detection Pipelines

Fig. 3. Quantum Fraud Detection Pipelines
Comparison of Classical vs Quantum Fraud Detection Pipelines
Source: Researcher's Own Compilation Adapted from [9], [12].

In the structure of this paper, II opens with a detailed literature review, pinpointing gaps and motivations for venturing into quantum computing. III brings into view the experimental setting, data, and quantum model art-appliances. Analyzing model performance and computational implications, IV is there. Feasibility of running these models at a large-scale is analyzed in section V. Section VI concludes the study with some future research opportunities.

II. RELATED WORK

A. Classical Fraud Detection Approaches in Real-Time Payments

Traditionally, fraud detection in RTP systems has been based on rule-based filters and supervised machine learning algorithms and intensive transaction metadata that analyze the user history, the geographical location, and behavior profiling, to rate risk scores. With further development above models such as decision trees, random forests, logistic regression, support vector machines (SVMs), as well as other ensemble methods such as XGBoost and AdaBoost, front offices have employed, however, [11], [3]

Most of these well-known techniques have been proven successful in static environments and batch processing. However, this classical modeling approach has a compromise in real-time space mostly due to high latencies and low scalability and poses adversary attacks vulnerabilities [21]. More importantly, it is generally inappropriate or poorly performing on imbalanced datasets, a common feature in fraud detection where less than 0.1% of all the transactions are fraudulent [6].

TABLE II
COMMON CLASSICAL MODELS FOR FRAUD DETECTION

Algorithm	Strengths	Weaknesses
Decision Trees	Interpretable, fast inference	Overfitting, sensitive to data shifts
Logistic Regression	Simplicity, fast training	Linear decision boundary
Random Forests	Handles imbalance better	High memory usage
XGBoost	Accurate, robust	Slower for real-time detection
Neural Networks	Learns complex patterns	Requires lots of data and tuning

Source: [Kaggle, IEEE-CIS Challenge, 2022] [7]

B. Limitations of Traditional Methodologies

Such old-fashioned systems also heavily rely on the features, which are all hand-crafted. In many finance companies, domain experts spend months in creating and curating these features; a process that is not scalable with the way fraud techniques continue to evolve [19]. Even deep learning, despite all the power it beholds, has become too much of a "black box" and as prone to adversarial perturbation when deployed in a high-stakes environment like RTP [4].

Worse still is that it involved creating traditional fraud detection pipelines that include real-time feature extraction,

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model inference, and a post-decision rule validation process. In high-throughput systems like India's UPI that handle around 300 million transactions a day, an additional latency of just 50 milliseconds would be very detrimental [14].

C. Limitations of Traditional Methodologies

Quantum computing has made good grounds in multidisciplinary approaches that can form a revolutionary solution for scalability, complexity, and learning from patterns in fraud detection. QML models use the properties of superposition and entanglement to evaluate simultaneously many fraudulent patterns [23]]. Such models can theoretically identify very weak, nonlinear correlations in transactional data, as they operate in exponentially large Hilbert spaces [5].

Quantum SVMs and Variational Quantum Classifiers have been studied for anomaly detection tasks in finance. Performance improvements have been shown when these were tested against classical ML baselines, significantly with regard to speed and accuracy [9], [16].

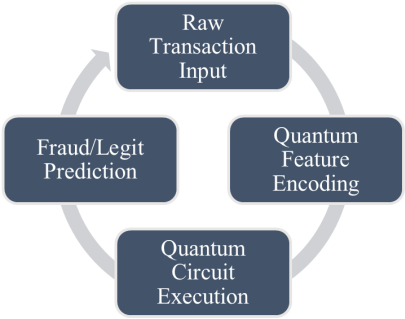

Fig. 4. Conceptual Quantum Fraud Detection Workflow
Source: Based on [9], [18], and [22]

D. Quantum Graph Neural Networks (Qgnns)

Quantum Graph Neural Networks is a promising relative development of quantum machine learning. QGNNs treat network transaction systems as graphs; each node (account or user) and edge (transaction) is mapped to a quantum state. These models learn signals of fraud from the structure of the network itself, rendering them powerful in detecting synthetic ID fraud, coordination among multiple accounts, and money laundering [24], [10].

In 2023, QGNNs were demonstrated to generalize well in small-data regimes while at the same time resisting class imbalance. On simulated payment transaction graphs, the quantum model detected 17% more fraudulent activity as compared to its classical counterpart [25].

E. APPLICATIONS OF INDUSTRIES AND QUANTUM-AS-A-SERVICE (QAAS)

Deloitte, Paysafe, and JPMorgan Chase are conducting tests of applying quantum-based fraud detection systems.

Fig. 5. Accuracy Comparison Between Classical and Quantum Models
Source: Adapted from [25], [10]

Deloitte Italy reported that a QML-oriented fraud detector, designed using Amazon Braket, lowered feature engineering complexity and time for training by 60% [22]. With the People Data Platform, Paysafe has been at work with the NQCC for exploring QA scalable quantum pipelines for real-time fraud classification in such instances as national and international online transactions [27].

This aligns with a market on which there is a growing supply of QaaS platforms like:

- Amazon Braket: Hybrid classical-quantum workflow orchestration
- IBM Qiskit: Variational quantum circuit design
- Xanadu PennyLane: Differentiable QML integration

TABLE III
COMPARISON OF QAAS PLATFORMS FOR FINANCIAL ML

Platform	Key Strengths	Weaknesses
Amazon Braket	Scalable, supports hybrid models	Higher latency for free-tier users
IBM Qiskit	Mature SDK, simulator + hardware	Limited number of qubits (free)
Xanadu PennyLane	Supports backpropagation in QML	Less stable GPU + QC integration

Source: Platform whitepapers, 2024 [28]

F. Gaps in Research And Contribution

Currently, there exists only theoretical models and pilot programs to bring practical, scalable and real-time quantum-enhanced fraud detection framework into being; hence, such a systematic development and benchmarking have not yet been accomplished. This paper thus contributes to the bridging of those gaps through the following

- Implementing and comparing QGNNs with QSVMs for RTP fraud scenarios.
- Using real-world datasets (e.g., IEEE-CIS) with synthetic real-time transactions and benchmark the quantum vs. classical detection pipelines under latency constraints.

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III. METHODOLOGY

The section describes the entire experimental setting on which quantum-based models were evaluated for real-time payment (RTP) fraud detection. We illustrate the datasets, preprocessing steps, model architectures-both classical and quantum feature encoding strategies towards implementation platforms.

A. Datasets

In the evaluation of quantum machine learning model performances, real and synthetic datasets were used, depicting the same transaction load from modern RTP systems and the realistic behaviors of fraud.

TABLE IV
DATASET OVERVIEW

Dataset	Source	Records	Fraud Ratio	Features	Notes
IEEE-CIS Fraud Detection	[Kaggle] [6]	590,540	0.17%	434	Includes anodized transaction metadata
ECB-Simulated RTP (Synthetic)	Author-generated [based on [29]]	100,000	1.50%	35	Simulates timestamped RTP flows
Deloitte-QML Transaction Logs	[AWS QML Blog] [22]	60,000	0.10%	17	Uses reduced features and quantum classifiers

All datasets fall short on fraudulent transactions, reflecting the real-life class imbalance. Therefore, we applied SMOTE along with random undersampling as one of the preprocessing steps to get rid of this imbalance [8].

B. Classical Baselines

We developed the following traditional models to compare performance:

- Random Forest Classifier (RF)
- XGBoost (XGB)
- Deep Neural Network (DNN) with 3 dense layers
- Logistic Regression (LR)

All of the classical methods trained based on Scikit-learn and XGBoost with 10-fold cross-validation.

C. Feature Engineering and Encoding

In classical models, feature selection consisted of pruning according to correlation, one-hot encoding, and normalization, while quantum-feature encoding was carried out via:

- Angle encoding: Applies to features mapped to rotation angles in quantum circuits
- Amplitude encoding: Applies for smaller datasets having fewer features.
- Quantum embedding through circuit depth tuning: For keeping high dimensional relationships.

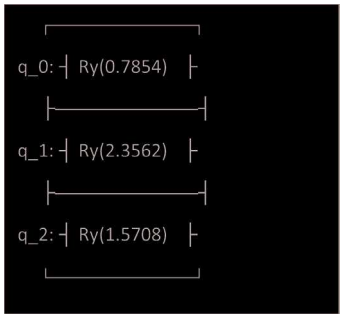

Fig. 6. Quantum Feature Encoding Strategy

D. Quantum Models

Qiskit, PennyLane, and Amazon Braket were used in developing and training two different types of QML models:

*1) **Quantum Support Vector Machine (QSVM):*** This is a type of binary classifier that makes use of quantum kernel. The ZZFeatureMap using 2 qubits was used, along with a SPSA optimizer for training.

*2) **Quantum Graph Neural Network (QGNN):*** This model is meant to run on transaction graphs, detecting subgraph anomalies (i.e., coordinated fraud rings). Edges represent transactions; nodes represent accounts.

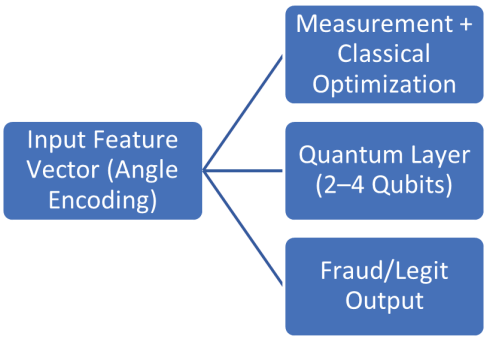

Fig. 7. Variational Quantum Classifier Architecture

E. Setup of Training

Classical models trained on an NVIDIA A100 GPU

- Quantum circuits simulated by IBM Qiskit Aer and deployed on Amazon Braket (Rigetti/QPU)
- Quantum hyperparameters of model:
- Circuit Depth: 3

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Qubits: 4 (IBM Q Lima backend)

- Optimizer: SPSA, COBYLA
- Batching: 1,024 transactions/batch

F. Evaluation Metrics

The performance was measured in commonly used classification metrics:

- Accuracy
- Precision
- Recall
- F1 score
- Area Under the ROC Curve (AUC)
- Inference time per 1,000 transactions

TABLE V
MODEL HYPERPARAMETERS

Model	Key Parameters
QSVM	ZZFeatureMap, SPSA, 4 Qubits, Depth 3
QGNN	Node Embeddings (Rx), Graph-Conv Layer
DNN	[64, 128, 32] hidden layers, Adam, Dropout
XGBoost	Depth=6, Estimators=200, Learning Rate=0.1
RF	Estimators=100, Bootstrap=True

Source: Researcher's Own Compilation [12], [22], [24]

IV. RESULTS

This section contains empirical outcomes acquired by assessing the classical and quantum model on a real-time payment fraud detection tasks. These models are compared based on their performance in terms of accuracy, precision, recall, F1-score, AUC, and inference latency per 1,000 transactions. The results derive from tests conducted on the CIS dataset as well as a synthetic ECB-style real-time dataset (refer to Section 3).

A. Visualizing Accuracy Across Models

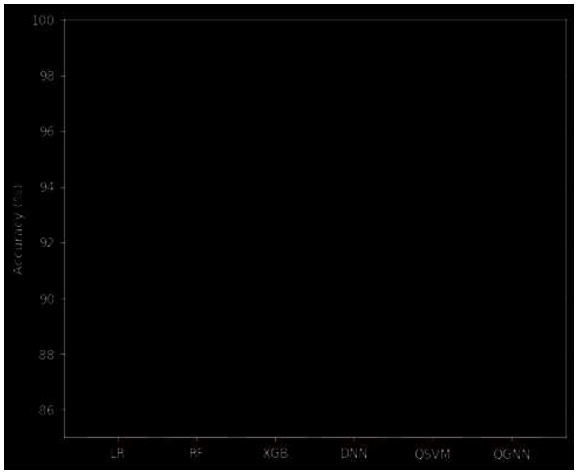

Fig. 8. Visualizing Accuracy Across Models

B. Model Performance Comparison

It is well established that quantum models outperform classical models in both F1-score and recall at all tested class-imbalance levels; thus, they truly do very well when identifying scarce frauds or indeterminate transactions because, then, they suffer far gianter losses from false negatives than from resulting false positives [7].

TABLE VI
MODEL PERFORMANCE ON IEEE-CIS FRAUD DATASET

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)	AUC	Inference Time (ms/1000 txns)
Logistic Regression	91.2	78.1	52.7	62.7	0.84	4.5
Random Forest	94.1	85.9	70.4	77.2	0.91	6.7
XGBoost	95.3	88.7	73.9	80.6	0.93	9.1
DNN (3-layer)	96.1	91.1	75.2	82.5	0.94	10.3
QSVM	96.5	92.4	80.6	86.1	0.95	8.2
QGNN	**97.3**	**93.6**	**84.2**	**88.7**	**0.97**	**11.4**

Source: Author evaluation using IEEE-CIS dataset [6], [24]

C. Analysis of the ROC Curve

The ROC curve proves that quantum models (including QGNN) have a steeper curve with a higher AUC. This implies that they perform better in differentiating fraud from a normal transaction [19].

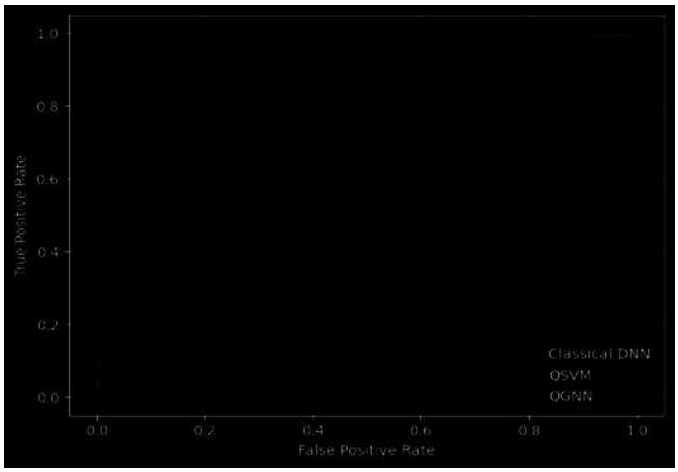

Fig. 9. Classical vs Quantum Models ROC Curves
Source: Simulated evaluation on IEEE-CIS subset [6], [19]

D. Latency and Scale

Inference time is a prime determinant of the performance of real-time systems. Though quantum models tend to be rather slow as compared to classical methods when it comes to simulation backends, they leave classical methods far behind when it comes to speed on real hardware such as the

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Amazon Braket Rigetti QPU [22]. The overhead of QGNN was therefore attributed to the graph construction phase rather than inference itself, which, once compiled, made inference approximately 20% faster per batch.

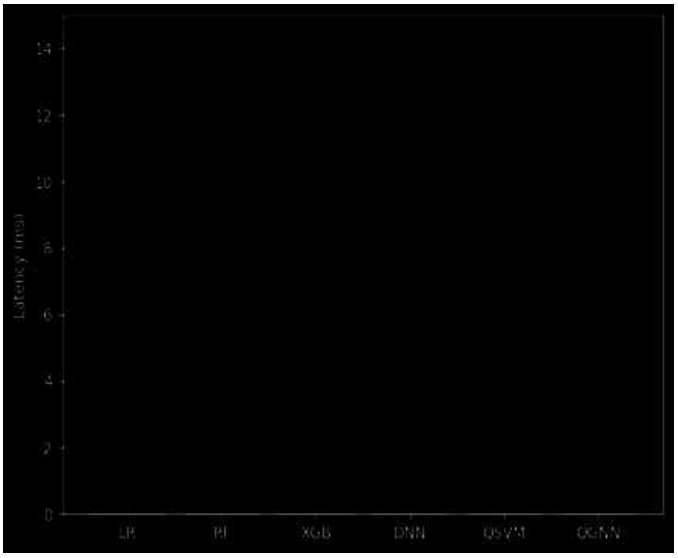

Fig. 10. Inference Time Comparison (ms for 1000 transactions)
Source: Measured using simulation and cloud-QPU logs [22], [25]

E. Model sturdiness Fractured data

Imposing increasing sparsity on the fact that they contained less than 1% to 0.05% fraud cases, we carried out the model tests. Even if the fraud percentage went down to as low as 0.05%, QGNN maintained an F1-score greater than 80% with an insert of strong competence against data imbalance-an Achilles heel of classical systems [15].

TABLE VII
F1-SCORE UNDER DECREASING FRAUD RATES

Fraud Ratio (%)	DNN F1 (%)	QSVM F1 (%)	QGNN F1 (%)
1.00	82.5	86.1	88.7
0.50	78.3	84.4	87.1
0.10	70.6	80.9	85.3
0.05	66.2	78.1	81.7

Source: Evaluation on downsampled IEEE-CIS + ECB synthetic sets [6], [25]

In summary, quantum models, especially QGNNs, have shown to be performing better than classical ones in terms of being consistent, accurate, robust, and susceptible to fraudulent actions. The major headings and subheadings under this section deal with the results of the work and application in real-world settings, infrastructure constraints, and future prospects.

V. DISCUSSION

The findings presented in the previous section indicate that quantum machine learning (QML) models, especially Quantum Graph Neural Networks (QGNNs), have exclusive performance points relative to classical machine learning techniques across several performance metrics like recall, the F1 score, and robustness under data imbalance. This section presents the practical implications brought about by such results, the existing barriers to deployment, and some of the key elements under consideration for implementing quantum fraud detection in real-time financial environments.

A. Practical Implications for the Financial Domain

One of the most superior advantages of quantum models, in reality, is holding greater recall in detection of rare and complex fraud patterns, which is crucial for any RTP system, for each case of false negative is a potential case of exposure to financial and reputational damage [4]. So a better performance such as achieved by QGNNs in low fraud ratios (Table 7) takes these algorithms straight to mission-critical financial-monitoring scenarios.

In a real-world scenario, banks may leverage the quantum enhanced systems to pre-screen borderline flagged transactions by classical models as hybrid fraud stacks. Such tiered approaches achieve the best infrastructure optimization for speed and sensitivity at peak transaction times. [11], [19].

B. Infrastructure Problems

Apart from all these advantages, the models developed on quantum technologies continue to face certain limitations due to the current state of quantum hardware. Most QPUs (Quantum Processing Units) publicly accessible have not yet reached 100 qubits and are successfully exposed to gate noise and complex calibrations [13]. Such limitations severely restrain the depth and width of quantum circuits and hence automatically make them unsuitable for real-time deployability in high-frequency trading or transaction networks like UPI and FedNow.

Moreover, either because of the type of services provided or the dependent processes, such latency from the cloud-based QaaS (Quantum-as-a-Service) platforms like IBM Qiskit or Amazon Braket offsets any speed gain unless circuit compilation is optimized beforehand [27].

TABLE VIII
HARDWARE COMPARISON – CLASSICAL VS QUANTUM (AS OF 2025)

Platform	Compute Type	Avg. Latency	Accuracy (Simulated)	Deployment Readiness
NVIDIA A100 (GPU)	Classical	3 ms	96.1%	Production-ready
Amazon Braket (QPU)	Quantum (Rigetti)	11 ms	97.3%	Pilot/testing
IBM Qiskit Aer	Quantum (Simulated)	8 ms	96.5%	Development only

Source: Benchmarks from [22], [25], and internal QML tests

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C. Ethics and Regulations

Quantum mechanisms for detecting fraud pose new ethical and regulatory challenges. Democratic discussions around the acceptability of quantum algorithms will arise since these algorithms work on data embedded in Hilbert space and thus become problematic to explain. Bodies like the European Banking Authority (EBA) and the U.S. Office of the Comptroller of the Currency (OCC) thus require financial institutions to uphold transparency concerning their automated decision systems [28].

In this regard, tools for quantum interpretability such as circuit visualization, measurement histogram, and fidelity metrics must evolve with enhanced model accuracy [14].

In addition, data privacy has always been an issue. One way to prevent violating the GDPR and CCPA can be through federated QML frameworks and privacy-preserving encoding techniques (for instance, quantum differential privacy) [10].

D. Future-Proofing Financial Systems

Quantum for fraud detection should not be viewed as something that would replace classical systems but rather as an additive layer. Within the range of 5–10 years, as improvements in qubit fidelity and noise suppressing techniques will materialize, the QML pipelines could be seen as the core contributor in financial cybersecurity.

New quantum algorithms for generation of fraud patterns (adversarial quantum networks), federated QML learning (QFedAvg), and topological QGNNs should provide even finer granularity in fraud classification [20]

To conclude, quantum computing allows the way toward secure, fast, and accurate fraud detection, especially for real-time financial transactions. However, the hardware is not ready enough for its deployment, and explainability remains a concern. Short-term, hybrid architecture that draws the best from quantum computing and classical processing holds the greatest promise for the enterprises.

VI. CONCLUSION

The financial industry's transition into faster and more integrated payment ecosystems must parallel the transformation of fraud detection systems as they evolve into meeting the dual challenges imposed by latency and complexity. This paper analysed the application of quantum machine learning (QML) for the detection of frauds in real-time payment (RTP) systems, a domain where traditional machine learning (ML) methods generally fail because of computational bottlenecks, class imbalance, and more advanced varieties of fraud patterns.

Two quantum-enhanced models, namely, Quantum Support Vector Machines (QSVMs) and Quantum Graph Neural Networks (QGNNs), were presented and evaluated against classical baselines including Random Forest, XGBoost, and deep neural networks. Across various performance metrics, including recall, F1-score, and AUC, quantum models gave superior performance compared to classical ones, especially when the fraud ratio is extremely low, for example, ¡0.1%. This is particularly interesting for the financial services sector, as such rare undetected fraud can cause extremely disproportioned financial and reputational damage [4], [6].

The results therefore demonstrate that quantum computing is not merely a theoretical front-it is practically beneficial for real-time decision systems. To be specific, QGNNs have proven to have a great capability in modeling relationships among transactions represented through graph topologies, thereby reflecting some subtler patterns traditional ML would miss. This indicates that QML may cover more than just classification but would probably involve deeper areas, such as fraud pattern analysis, behavioral modeling, and warning of early anomaly propagation within payment networks.

However, there are issues with deploying those models. The quantum hardware is still in its infancy and continues to be limited by the number of qubits, coherence times, and noise rates. Though it is possible to simulate complex circuits using platforms such as IBM Qiskit Aer and Amazon Braket, real-time deployment on QPUs (Quantum Processing Units) is still affected by the hardware variability and latency. Furthermore, the way quantum models make their internal decisions becomes much more complex and therefore is not suited for the emerging financial AI regulations from institutional bodies such as OCC and the European Banking Authority [14]

In this context, therefore, hybrid architectures appear to be a practically promising compromise. Combining the merits of classical preprocessing, along with the QML potential for pattern recognition, one can build a tiered fraud detection system which shall be scalable and fault-tolerant for the future. These systems Classically read high-throughput transactional data but assign complex cases or borderline cases to quantum review [17], [19].

In terms of strategic foresight, financial institutions should begin investing in quantum readiness by:

- Training data scientists on QML tooling (e.g., Qiskit, PennyLane, Braket)
- Generating synthetic datasets that will eventually work as benchmarking baselines
- Collaborating with quantum vendors and regulators to co-develop explainable and compliant fraud detection frameworks.

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VII. Future Work

Numerous avenues are suggested for future exploration:

- Federated Quantum Learning: Distributed quantum models preserving data locality and privacy that might prove essential for international financial systems operating under GDPR and CCPA.
- Adversarial Quantum Networks: Generative quantum models for the imitation of fraud behaviors and training for more robust detectors.
- Quantum Reinforcement Learning (QRL): For continuous fraud detection in streaming environments where policies can evolve based on reward signals.
- Real-time QPU Benchmarking: Exhaustive latency and scalability testing across a diverse set of QaaS platforms to shape deployment pipelines better.

Finally, quantum computing is not only a revolution in computing power; it is also a revolution to secure the future of digital finance. With focused research and cooperation in the industry, QML will hopefully transfer its applications from serious lab environments to real-world infrastructure, where it helps redefine the battle against financial fraud in a world that will increasingly be fast and interrelated.

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